Suitable habitats shifting toward human-dominated landscapes of Asian elephants in China

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Abstract

Although intensifying human activity in Asian elephants' natural habitats has led to gradual habitat changes, fragmentation, and contraction in recent decades, their population has continued to increase and disperse due to a series of conservation efforts, resulting in increased severe human-elephant conflicts. The habitat adaptation strategies of Asian elephants remain unclear. Here, we used the maximum entropy models to explore habitat selection strategies of Asian elephants at different spatial scales in Southwest China based on the occurrence data. Using habitat suitability predictions, we identified the key environmental, anthropogenic, and climatic variables influencing Asian elephants' habitat utilization. We also assessed the historical dispersal trend of Asian elephants and the overlap between suitable habitat ranges and human-dominated regions. The results showed that inherent topographic variables, such as elevation and slope, strongly influence the habitat selection of Asian elephants across spatial scales and that human activity influence is stronger at larger scales. There is currently approximately 17,744 km2 of suitable habitat and 25,590 km2 of sub-suitable habitat for Asian elephants in China, mainly in Xishuangbanna and central and south Pu'er, with 63.73% of these habitats overlapping human-dominated regions. Under the high-emissions climate change scenario, suitable and sub-suitable habitats of Asian elephants would shrink by 61% and 31% by the 2050s. In areas of high overlaps between suitable habitats of Asian elephants and human-dominated regions, strategies such as habitat restoration, construction of ecological corridors, and land use improvement could effectively alleviate human-elephant conflict and promote human-elephant coexistence.

Introduction

From June to August 2021, a herd of Asian elephants in far-southwest China attracted unprecedented attention from the public because of its more than 500 days and 1,300 km journey from their original home range in Xishuangbanna to Kunming, the hosting city of the UN Biodiversity Conference (CBD COP 15) (Part 1) (CCTV.com, 2021). Campos-Arceiz et al. (2022) and Jiang et al. (2021) deemed this journey as an attempt at dispersal, which is a general behavior of animals (Debeffe et al., 2012; Gosselink et al., 2010; Loe et al., 2010; Nunes et al., 1996), and involves individuals or propagules leaving their original habitat to establish new habitat in other areas or join a new population (Lidicker, 1975; Ronce, 2007; Stenseth & Lidicker, 1992; Stieha et al., 2014). In Asian elephants, range expansions by dispersal or recolonization occur frequently and widely in many countries (Baskaran et al., 2011; Gubbi et al., 2014; Neupane et al., 2020; Sivasubramanian & Ramakrishnan, 2021). In China, the range of the Asian elephant population expanded from the Mengyang Nature Reserve, Xishuangbanna National Nature Reserve (hereafter Mengyang Reserve) and gradually increased northward in the 1990s (Wu et al., 1999; Xiong, 1996; Yang et al., 1987; Yu et al., 2015).

The exodus of Asian elephants from reserves is driven by many factors, including but not limited to vegetation coverage increase, habitat fragmentation, and population growth (Bowler & Benton, 2010; Koirala, 2016; Neupane et al., 2020; Talukdar et al., 2020; Taher et al., 2021). In China, the original habitats of Asian elephants in nature reserves have been strictly protected and managed by local nature reserve management bureaus under the Regulations of the State Council with little interference from human activities since the 1950s, and canopy cover has continued to increase (Chen et al., 2006; Liu et al., 2017). As a result, the forest understory food sources and accessibility for elephants were reduced (Campos-Arceiz et al., 2021; Liu et al., 2020). On the other hand, the extensive economic and food crops outside the reserves appeal to elephants due to ease of accessibility (Dai, 2017; Linkie et al., 2007; Meijaard & Erik, 2013). Taken together, these factors have facilitated the dispersal of elephants from relatively intact forests in reserves to human-dominated regions with agricultural land and forestry ecotones (de la Torre et al., 2021; Huang et al., 2019; Sivasubramanian & Ramakrishnan, 2021).

Climate change has also been proposed as a key driver of range changes for wildlife (Bai et al., 2022). Many studies indicate that climate change affects species distribution and behavior (Cohen et al., 2018; Dai et al., 2019; Root et al., 2003), damages ecosystems and biodiversity (Parmesan & Yohe, 2003; Sinervo et al., 2010), endangers species' survival, and increases species susceptibility to extinction (Alamgir et al., 2015; Foden et al., 2013; Mumby et al., 2019). Changes in temperature and precipitation regimes could affect species' geographical distribution through vegetation and food distribution, which may also increase their potential distribution area (Kriticos et al., 2003; Peng et al., 2002; Wuethrich, 2000). Wang et al. (2021a) proposed that extreme dry heat caused a dramatic degradation of herbaceous plants and shrubs, leading to the northward migration of Asian elephants in China in 2020.

In recent years, the overlap between Asian elephant ranges and human-dominated regions has continued to increase. Such anthropogenic interference has intensified in Asian elephant habitats, and frequent human-elephant conflicts have occurred (de la Torre et al., 2021; Ekanayaka et al., 2011; Sukumar, 1990; Wilson et al., 2015). In China, the range of the 293 Asian elephants is limited to three prefectures in the Southwest, in Yunnan Province (Yunnan Provincial Forestry and Grassland Administration, unpublished data). Nevertheless, China’s Asian elephant population has continued to increase and disperse, leading to increased activities in human-dominated regions (Huang et al., 2019; Tang et al., 2020; Wang et al., 2021b; Yu et al., 2015). As a result, the frequency of human-elephant conflicts has increased in the past decade, resulting in dozens of injuries or deaths and a cumulative compensation of 173 million yuan for Asian elephant hurts and damages (CCTV.com, 2022; Zhang et al., 2022). In the face of increasingly severe human-elephant conflicts, understanding the movement and habitat of Asian elephants is urgent. The knowledge of Asian elephant habitat in human-modified landscapes remains extremely poor, especially their habitat selection strategy and process of range change. The information is crucial for managing human-elephant conflicts. Here, we retraced the dispersal history routes of the Asian elephant to predict dispersal trends, applied the maximum entropy model (MaxEnt) and monitoring data to investigate the habitat selection strategies and habitat suitability of Asian elephants. We examined the following hypotheses: 1) the suitable habitats of Asian elephants are shifting to human-dominated regions which provide food resources and shelters; 2) because Asian elephants inhabiting human-dominated regions have a frequent interaction with humans, anthropogenic interference variables are more important for their habitat selection.

Materials And Methods

Study area and animals

A total of 293 Asian elephants distributed in nine counties of three prefectures in Yunnan Province, Southwest China (Xishuangbanna, Pu'er, and Lincang) were divided into four populations (Fig. 1). We selected the Pu'er-Banna population, which is the largest in China (accounting for 64.5%), and whose original range is Mengyang Reserve, Jinghong, Xishuangbanna (Yunnan Provincial Forestry and Grassland Administration, unpublished data), as our study animals. Over the past three decades, the Pu'er-Banna population frequently traveled between Pu'er and Xishuangbanna, to the surrounding areas. Currently, the population is distributed in five counties, including Jinghong (Xishuangbanna), Simao, Jiangcheng, Ning'er, and Jinggu (Pu'er) (He, 2013; Yu et al., 2015; Zhu et al., 2019). Our study area, Yunnan Province in Southwest China (21°10'-26°22' N, 99°09'-104°16' E), focused on the range of the Pu'er-Banna population, including the areas journeyed by the northward-moving herd in 2021, totaling approximately 128,723 km^2 (Fig. 1). The elevation is between 76 and 4,248 meters above sea level. The study area is characterized by tropical and subtropical climates, with distinct dry and wet seasons. The vegetation is dominated by subtropical monsoon evergreen broad-leaved forests (Jiang, 1980). The resident human population in the five counties is 1.6 million (People's Government of Pu'er City, 2022; People's Government of Xishuangbanna Dai Autonomous Prefectural, 2021).

Presence points

In this study, we obtained 10,905 presence points of Asian elephants from the Asian elephant tracking and monitoring by local rangers equipped with UAVs during 2019–2021 in collaboration with the local forestry and grassland department (Fig. 1). One to two presence points per day per individual or herd were recorded. We also integrated the data captured at 87 infrared camera trapping sites in Mengyang Reserve into the presence points. According to previous observations of the movement of Asian elephants in the study area, the maximum daily movement distance is approximately 10 km, so three scales of 1×1, 5×5, and 10×10 km grids are used in our study to determine the importance of each variable at different scales to habitat selection for Asian elephants. The 1×1 km grid with higher accuracy was used to analyze the suitability and distribution of Asian elephant habitat. We then filtered the presence points of Asian elephants by randomly choosing one locality within the 1×1, 5×5, and 10×10 km grids, respectively, to reduce the spatial autocorrelation caused by the proximity of the presence points.

Environmental and human variables

To assess the suitability and selection of the Asian elephant habitat at different scales, we selected 15 critical environmental and anthropogenic variables based on previous studies (Jiang, 2019; Neupane et al., 2020; Wadey, 2018). The environmental variables included elevation, slope, aspect, roughness, patch density, normalized difference vegetation index (NDVI), river density, and distance to rivers, while the anthropogenic variables included human modification, land use, nighttime lights, road density, distance to roads, settlement density, and distance to settlements. Detailed sources and processing methods are provided in Appendix. The definitions of land use and vegetation are provided in Appendix.

We obtained 19 bioclimatic data from the WorldClim database versions 1.4 and 2.1 to represent climatic variables of past and present periods (Fick & Hijmans, 2017). The future periods were represented by bioclimatic data for the 2041–2060 period under the representative concentration pathway (RCP) 8.5 which predicts global average temperature will rise by 1.4–2.6°C in the 2050s (IPCC, 2019; Moss et al., 2008). Due to the long-term and large-scale impacts of climate, animals with climate niche conservatism have more passive adaptation to climate change than active selection (Sales et al., 2017). Therefore, the effects of climate variables on the habitat selection of Asian elephants were excluded from this study. Then, we used three periods of bioclimatic data with 30" (~ 1 km) spatial resolution to analyze the change in the distribution of Asian elephants in recent decades (details listed in Appendix).

To avoid the effect of collinearity of variables on the accuracy of the MaxEnt model, only uncorrelated variables (Spearman correlation absolute values < 0.8) were retained (Yackulic et al., 2012). Consequently, we removed roughness and retained bio1, bio3, bio7, bio13, bio14, and bio15 from the 19 bioclimatic variables in the past and present periods and bio1, bio3, bio7, bio13, bio14, bio15, and bio17 in the future period. All spatial data were resampled to 1, 5, and 10 km resolutions. The spatial data were processed in ArcGIS 10.6.

MaxEnt modeling

The maximum entropy algorithm was used to model the distribution of Asian elephants (Alamgir et al., 2015; Htet et al., 2021; Wang et al. 2018). The presence points of Asian elephants collected from August 2019 to August 2021 as occurrence data, integrated with climatic, environmental, and anthropogenic interference variables, were substituted into the MaxEnt model to evaluate the suitability and selection of habitat of Asian elephants and to analyze the distribution changes in the climate-suitable region of Asian elephants in recent decades.

In the MaxEnt model setting, 75% of occurrence data were randomly selected as the training set, the remaining 25% as the test set for model validation, and other settings were kept as recommended default. To ensure the stability of the model results, we ran 20 replicates with bootstrap type. Variable contributions and response curves were generated to analyze the relative importance of the variables and their impact on the prediction results (Li et al., 2019; Yang et al., 2020). The prediction results of the model were evaluated by the area under the curve (AUC) of the receiver operating characteristic curve (ROC) (Phillips et al., 2004). The MaxEnt model outputs the average habitat suitability index (HSI) in logistic regression to assess the Asian elephant habitat in the study area. We classified habitat into suitable habitat (0.376–0.853), sub-suitable habitat (0.133–0.376), and unsuitable habitat (0-0.133) by Jenks based on HSI. We analyzed the changes in the distribution of climate-suitable regions of Asian elephants in three periods in the same way. Climate regions were distinguished into the suitable region (0.366–0.738), sub-suitable region (0.110–0.366), and unsuitable region (0-0.110) by Jenks. The area changes in the three periods were calculated to reflect the dynamic change in the climate-suitable region of Asian elephants in recent decades.

Analysis of dispersal history

We obtained historical occurrence records of Asian elephants from the Pu'er Forestry and Grassland Bureau and combined them with those of Yu et al. (2015). We processed the occurrence records to acquire their dispersal history routes in Pu'er in the 1990s, 2000s, 2010s, and 2020s. Based on the corridor width of Asian elephants (Lin et al., 2008b; Osborn & Parker, 2003; Ravisankar et al., 2019) and previous observations of their movement, we compared the area of the 5 km buffer zone along the Asian elephant dispersal route in 4 periods and the original habitat in the Mengyang Reserve to grasp the dispersal process. Then, we determined if there was a tendency for Asian elephants to disperse into human-dominated regions by measuring area changes in cultivated land in the 5 km buffer zone.

Analysis of habitat overlap

The extent of overlap between the suitable habitat of the Asian elephant and cultivated land could reflect their dependence on crops. To obtain the distribution of overlap, we focused on the relationship between Asian elephant habitats and human-dominated regions, which often promotes high rates of human-elephant conflict (La Due et al., 2021). The cultivated lands and a 1 km buffer zone in settlements and roads were defined as human-dominated regions based on previous studies (Baloch et al., 2017; Naha et al., 2019) and actual investigations. We then superimposed the suitable habitats onto the human-dominated regions to estimate the extent and distribution of overlap.

Results

The dispersal trend of the Asian elephants

Based on the sources and time of Asian elephants’ dispersal to Pu'er, we divided them into - Herds to describe the dispersal process (Fig. 2). In the early 1990s, Herd , the first dispersal from Xishuangbanna, arrived in western Pu'er and then traveled back and forth between Pu'er and Xishuangbanna (W1). Herd dispersed to the central part of Pu'er and formed a resident herd in 1995 (M1). In the 2000s, Herd continued to move northward and settled in 2006 (M2). At the same time, Herd dispersed to the border between Pu’er and Xishuangbanna and formed a resident herd (M3). Herd was separated due to the construction of the hydropower station and formed an independent resident population (W1) and then continued to move westward after 2007 (W2, W3). Most of the dispersal of Asian elephants occurred in the 2010s. In 2010, Herd dispersed and formed the third resident herd in the middle part (M4). Then, the first dispersal of Herd established a new dispersal route (E1) in 2011 and left for the north to form a resident herd in the next year (E2). Later, Herd returned after several northward dispersals in the late 2010s (E3, E4). By 2020, the northward-moving herd had dispersed to northeast Pu’er for the first time (E5). At this point, Asian elephants active in Xishuangbanna in the early years gradually formed the current distribution through decades of exploration and dispersal (Fig. 2).

The accumulative area of the 5-km buffer zone along the Asian elephant dispersal route increased from 719.37 km^2 to 9,730.02 km^2 (1352.6%) in the past 30 years, of which the cultivated land increased from 54.28 km^2 to 1,074.42 km^2 (1979.4%) (Fig. 3). The increasing area of the dispersal region in four periods (the 1990s, 2000s, 2010s, and 2020s) showed that the distribution range expanded with increasing dispersal distance, which covered a larger area of the human-dominated regions represented by cultivated lands. The results also showed that the dispersal process of Asian elephants is gradually accelerating, with a large number of dispersal activities occurring in the 2010s, and the area of the dispersal region increased by 6.44 times that of the previous decade (Figs. 2 & 3).

Habitat selection of the Asian elephants

After spatial filtering, we obtained 1530, 188, and 71 presence points at 1, 5, and 10-km scales for the MaxEnt model. The average training AUCs for the replicate runs were 0.881, 0.938, and 0.927, and the standard deviations were 0.003, 0.007, and 0.009. Among the 15 variables included in the MaxEnt model on three scales, the top six relative contributions are shown in Table 1. All the total relative contributions of the top six variables on the three scales were more than 80%, which was the main factor affecting the habitat selection of Asian elephants. See Appendix for the complete contribution table and the response curves of the top six variables.

Table 1
The top six relative contributions on the 1, 5, and 10-km scales

Variables at a 1-km scale	Contribution (%)	Variables at a 5-km scale	Contribution (%)	Variables at a 10-km scale	Contribution (%)
elevation	66.9	elevation	45.6	Elevation	28
slope	9.8	human modification	8.9	Human modification	14.4
NDVI	8.3	slope	7.5	Road density	13.8
vegetation	6	NDVI	6.9	Vegetation	9.3
human modification	3	settlement density	6.6	Aspect	8.4
landuse	1.8	aspect	4.9	Settlement density	7.4

Based on the response curves of the six dominant variables at three scales, we found that Asian elephants have different habitat selection strategies at different spatial scales. The most suitable habitat at the 1 km scale was characterized by 1) elevation between 750-1,200 m, 2) slope less than five degrees, 3) NDVI above 0.88, 4) a large number of crops and broad-leaf forests, 5) human modification between 0.13–0.22, and 6) cultivated land and forest. The most suitable habitat at the 5-km scale was characterized by 1) elevation between 750-1,200 m, 2) human modification less than 0.23, 3) slope less than six degrees, 4) NDVI approximately 0.9, 5) settlement density less than 0.07, and 6) aspect between

140–270°. The most suitable habitat at the 10 km scale was characterized by 1) elevation between 750-1,200 m, 2) human modification less than 0.22, 3) road density less than 0.3, 4) large numbers of broad-leaf forests, 5) aspect between 140–260°, and 6) settlement density less than 0.07. In general, the influence of anthropogenic interference variables on the habitat selection of Asian elephants was more crucial at a larger scale (Appendix).

Climate impacts on Asian elephant habitat

The average training AUCs for the replicate runs of past, present, and future periods in the MaxEnt model with bioclimatic variables were 0.925, 0.928, and 0.932 (all standard deviations are 0.001), respectively.

The MaxEnt model statistical area was 131,733 km^2 (based on the grids), the distribution of suitable regions, sub-suitable regions, and unsuitable regions in the past, present, and future periods are shown in Fig. 4, and the area and proportion are shown in Table 2.

Table 2
Area of Asian elephant distribution range divided by climate in the past, present, and future

Period	Suitable region(km^2)	Sub-suitable region(km^2)	Unsuitable region(km^2)
Past	8,010(6.08%)	16,188(12.29%)	107,535(81.63%)
Present	6,985(5.30%)	11,427(8.68%)	113,321(86.02%)
Future	6,937(5.27%)	9,881(7.50%)	114,915(87.23%)

The rank sum tests of pairwise combinations of climate suitability in the three periods showed that the difference in climate between the three periods was statistically significant ($P < 0.01$). Compared with the present distribution range, only 30.76% of the sub-suitable region and 61.15% of the suitable region remain stable. In recent decades, the suitable and sub-suitable regions of Asian elephants have shrunk and gradually transformed into unsuitable regions. The contraction and transformation trend will continue in the coming decades.

The habitat distribution of the Asian elephants

In this study, habitat suitability was assessed at the most detailed scale, 1 km, and distinguished by Jenks natural breaks method. The size of the study area was 128,229 km^2, of which the suitable habitat area was 17,744 km^2 (13.84%); the sub-suitable habitat area was 25,590 km^2 (19.96%); and the unsuitable habitat area was 84,895 km^2 (66.21%). In terms of habitat distribution, suitable and sub-suitable habitats were mainly found in Xishuangbanna and southern Pu'er, the original habitat of Asian elephants, which had a large area with habitat connection. In addition, suitable habitat was found in southeastern Xishuangbanna, central and northwestern Pu'er, and regions between the Wuliang Mountains and Ailao Mountains. Only some fragmented suitable habitats existed in the rest of the study area (Fig. 5).

The status of the habitat distribution of the Pu'er-Banna population showed that the existing habitat is currently better than any other potential habitats in terms of suitability, size, and integrity (Fig. 5). We also found that the existing habitat of Asian elephants failed to connect with the potential habitats.

The results of the superposition of Asian elephant habitat distribution and the layers related to human activity showed that the cultivated land area accounted for 31.87% of the total area of the study area and accounted for 26.84% of the original distribution ranges of Asian elephants (Fig. 6). Many human settlements and major roads were spread throughout the suitable habitat (Table 3). In the study area, the overlap area of the suitable and sub-suitable habitat of Asian elephants with the human-dominated regions was 27,617.27 km^2, accounting for 63.73%. Notably, 79.14% of the overlapping ranges (21,857.43 km^2) were in the original distribution region of Asian elephants. Therefore, the suitable habitats of Asian elephants highly overlapped with human-dominated regions (details shown in Table 3).

Table 3
The area of human-dominated regions within the Asian elephant habitat

Range	Suitable habitat	Sub-suitable habitat
Study area (km^2)	17,744	25,590
Road (km)	5,494.31	9,414.05
Settlement	1,442	2,719
Cultivated Land (km^2)	5,383	7,335
Overlap with human-dominated regions (km^2)	11,066.89	16,550.38

Discussion

In this study, we investigated the impact of human activities on the distribution and dispersal of Asian elephants by tracing their historical movement routes, analyzed the habitat selection strategy at different scales, and identified the habitat distribution and suitability. We found that the habitat of Asian elephants highly overlapped with the human-dominated regions. The increasing area of human-dominated regions during the dispersal process and habitat expansion of Asian elephants indicates that Asian elephants are increasingly utilizing and relying on human-dominated regions and benefiting from forest agriculture matrices despite the increasing frequency of interaction with humans causing elephants to be more vulnerable to anthropogenic interference (Leimgruber et al., 2003; Tang et al., 2020).

Dispersal of Asian elephants

As a category national key protected wild animals in China, the population of Asian elephants has increased in recent years (Li, 2009; Zhang et al., 2006; Yunnan Provincial Forestry and Grassland Administration, 2019, unpublished data). Furthermore, due to the implementation of the policy banning hunting, logging, and burning, vegetation recovery accelerated, and the forest canopy connection increased in the Mengyang reserve (Campos-Arceiz et al., 2021; Liu et al., 2020). Although the available data could not identify which factors dominate the dispersal of Asian elephants, population growth and forest canopy connection increase may be the main reasons for the dispersal. The dispersal of the Asian elephant has gradually intensified in recent decades, driven by a variety of factors (Wang et al., 2017; Zhu et al., 2019). The Asian elephant dispersal in the 2020s is less than before simply because the duration is short and the current dispersal is dominated by the northward-moving Asian elephant herd, so we expect the dispersal process and the area of the human-dominated regions to increase dramatically by the end of the 2020s. Based on the dispersal trend and dispersal precedents of the Pu'er-Banna Asian elephant population in recent decades, we predict that the population will disperse continuously, the distribution range would increase gradually with the increasing area of human-dominated regions in the future, and the greatest possibility of dispersal direction is still north.

Habitat selection of Asian elephants

Wildlife adapts to the particular combination of habitat features by habitat selection to meet their basic requirements for space, food, water, and cover (shelter) (Yarrow, 2009). Nevertheless, the native habitat loss of Asian elephants is widespread due to urban development, habitat encroachment, and expansion of agriculture (Chen et al., 2022; Mariati et al., 2014; Sarma et al., 2008), which resulted in the dispersal of populations to new habitats, the extension of habitat ranges and change in habitat selection strategies (Baskaran et al., 2011; Gubbi et al., 2014; Yu et al., 2015).

Our results suggest that the habitat selection of Asian elephants is influenced by the natural environment and anthropogenic interference variables. Although the importance of variables varies, the inherent topographic variables are dominant at different scales (Table 1). Most of the habitats of Asian elephants are gentle slopes or flat areas. Asian elephants avoid steep slopes or occasionally use "Z"-shaped routes to walk on steep slopes to save energy (Sach et al., 2019). The Asian elephants prefer to use habitats at elevations between 750-1,200 m in our study areas, although the elevation of the distribution range of Asian elephants is almost between 500-2,200 m (Appendix).

In addition to the inherent terrain variables, the influence of food is mainly reflected by the optimal foraging theory; that is, animals tend to acquire the highest food energy with the shortest feeding time (Pyke et al. 1977). Therefore, the lower availability of palatable wild food plants and the temptation of crops due to easy access, high nutrition, and good palatability drove Asian elephants to mostly be utilized in human-dominated regions (Sivasubramanian & Ramakrishnan, 2021), despite the utilization of habitats of elephants being based on a risk minimization strategy (Graham et al., 2009).

We also found that the variables of distance to rivers and river density have rare contributions to the model, especially at a larger spatial scale (Appendix). The reason may be the numerous rivers in the study area, so that water sources are not a limiting factor for Asian elephants at a larger spatial scale, rather than the distribution of water sources not being important. Similarly, the low contribution of distance variables at a larger scale could also be due to the dense distribution of rivers, roads, and settlements.

For shelter and shade, Asian elephants prefer broad-leaved forests with higher NDVI. Due to the preference for broad-leaved forests and Cultivated vegetation far more than other vegetation, high NDVI areas include not only forests but also farmlands and commercial forests with densely planted crops. The extensive cultivated lands in Asian elephant habitats are usually agroforestry ecotones with higher NDVI on small scales, which can meet the shelter and food requirements (Huang et al., 2019), which also explains why cultivated lands were suitable habitats for Asian elephants.

The influence of human activities on the habitat selection of Asian elephants can be reflected by the contribution of anthropogenic interference variables (Appendix). Most of the six dominant variables affecting the habitat selection of Asian elephants at three different scales were related to humans (Table 1). The results demonstrated that the impact at a small scale is mainly through human modification and patch type determined by vegetation and land use patterns. At a medium spatial scale, human modification and the distribution of settlements are more critical. At a larger spatial scale, variables related to human activities, such as human modification of the environment, vegetation, and the distribution of roads and settlements, affected the habitat selection of Asian elephants (Table 1). Such findings suggest that human impacts play more important roles at a larger spatial scale. Furthermore, our results demonstrate that although lower levels of human modification, road, and settlement density tend to be more suitable for Asian elephants, their dependence on human-dominated regions makes them less likely to choose areas with no human intervention.

In general, excluding the fixed natural variables such as elevation, slope, and aspect, the distribution of human-dominated regions and the human activity intensity become more important in the habitat selection of Asian elephants with the increase of spatial scale, and even a key factor in determining habitat suitability. It is also likely that Asian elephants are adapting to and becoming dependent on human-dominated regions.

Habitat distribution of Asian elephants

The habitat fragmentation and loss caused by agricultural and human development, illegal logging, and land use conversion have caused the disappearance of Asian elephants from 95% of their historical range (Leimgruber et al., 2003; Sitompul et al. 2013; Sukumar, 2006). The habitat distribution of Asian elephants is affected by various variables, such as terrain, vegetation, climate, and anthropogenic interference (Kanagaraj et al. 2019; Lin et al., 2008a; Pradhan & Wegge 2007).

In this study, the differences in the distribution of climate-suitable regions for Asian elephants on a temporal scale suggest that under the warmest climate change trend in the future, the gradual disappearance of sub-suitable regions may have an increasing impact on Asian elephants. If future climate conditions improve, the suitable regions for Asian elephants may remain stable or even gradually expand. Nevertheless, the contraction of the suitable and sub-suitable regions that gradually transformed into unsuitable regions was not consistent with the dispersal of Asian elephants. Therefore, climate variables are less likely to be the main factors driving changes in the habitat distribution of Asian elephants.

The overlap between suitable habitat of Asian elephants and human-dominated regions

The results of the study showed that the overlaps between suitable and sub-suitable habitats of Asian elephants and human-dominated regions were as high as 63.73%. While human disturbance and land-use change limit Asian elephant habitats (Sampson et al., 2019; Tripathy et al., 2021), linear infrastructures such as high-grade roadways also hinder Asian elephant movements (Blake et al., 2008; Dasgupta & Ghosh, 2015; Huang et al. 2020). The spatial overlap between human activities and elephant habitats would make human-elephant conflict more frequent (Wilson et al., 2015). Despite this, Asian elephants still profit from habitats that overlap with human-dominated regions. The terrain of human-dominated regions is relatively flat for the movement of Asian elephants. Cultivated land provides easily accessible and nutritious food for Asian elephants, and adjacent patches of small forest provide shelter for Asian elephants; consequently, the environment of the agroforestry ecotone becomes their ideal habitat. In addition, profiting from the wildlife protection policy in China, Asian elephants in human-dominated regions have essentially no direct hunting threat. The lower population density in Southwest China allows Asian elephants to tolerate relatively low anthropogenic interference (Xu, 2017). All these conditions lead to a high overlap of Asian elephant habitats with human-dominated regions.

Conclusions

In conclusion, our study has demonstrated that human activities have increasingly influenced Asian elephants' habitat distribution and selection. However, we did not find the dominant factors driving habitat change for Asian elephants. Recent studies have shown that clarifying the dominant factors driving the dispersal of Asian elephants will be the key to studying the behavioral changes of Asian elephants and mitigation of human-elephant conflicts (Bai et al., 2022; Campos-Arceiz et al., 2021; Wang et al., 2021b). At the same time, the establishment of ecological corridors is a practical measure to address the management and conservation challenges posed by the dispersal of Asian elephants (Jasmine et al., 2015; Menon, 2017; Shaffer et al., 2019), which could increase the exchange of individuals between habitat patches and promote genetic exchange (Hess & Fischer 2001; Tewksbury et al. 2002). Our study also revealed a high overlap between Asian elephant habitats and human-dominated regions; this is not likely to change in the short term, leading to changes in Asian elephant behavior, frequent human-elephant contact, increasing difficulty in managing and protecting Asian elephants, and threatening to the lives and property of residents (Liu et al., 2017; Neupane et al., 2020; Tang et al., 2020; Wilson et al., 2015). It is necessary to strengthen monitoring and early warning and guide the dispersal of Asian elephant populations to avoid settlements and main roads to protect and manage Asian elephants and alleviate human-elephant conflict.

Declarations

Author Contributions

X.J., X.L., Q.Y., and Z.H. involved in conceptualization and methodology. Q.Y., Z.H., and T.X. carried out the investigation. Q.Y. and Z.H. performed data analysis. Q.Y. wrote the manuscript. Q.Y., Z.H., C.H., and K.O. involved in reviewing the manuscript, and X.J. and X.L. involved in supervising the study and finalizing the manuscript.

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Figures

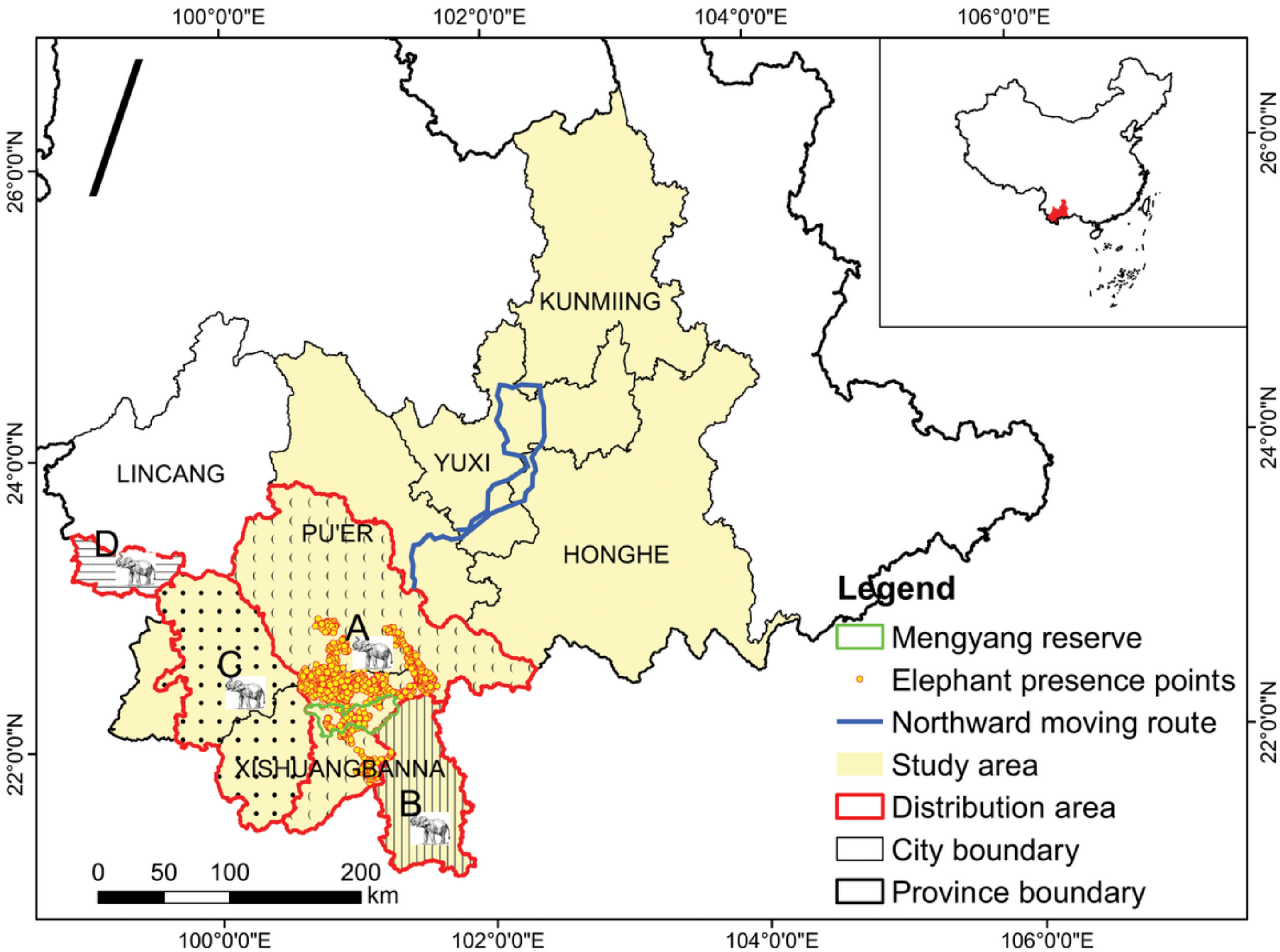

Figure 1

Study area with presence points of the Pu'er-Banna Asian elephant population and the northward moving route of the Asian elephant herd. The four Asian elephant icons depict the distribution of Asian elephant populations: (**A**) Pu'er-Banna population (**B**) Mengla population (**C**) Lancang-Menghai population (**D**) Cangyuan population (Yunnan Provincial Forestry and Grassland Administration, unpublished data)

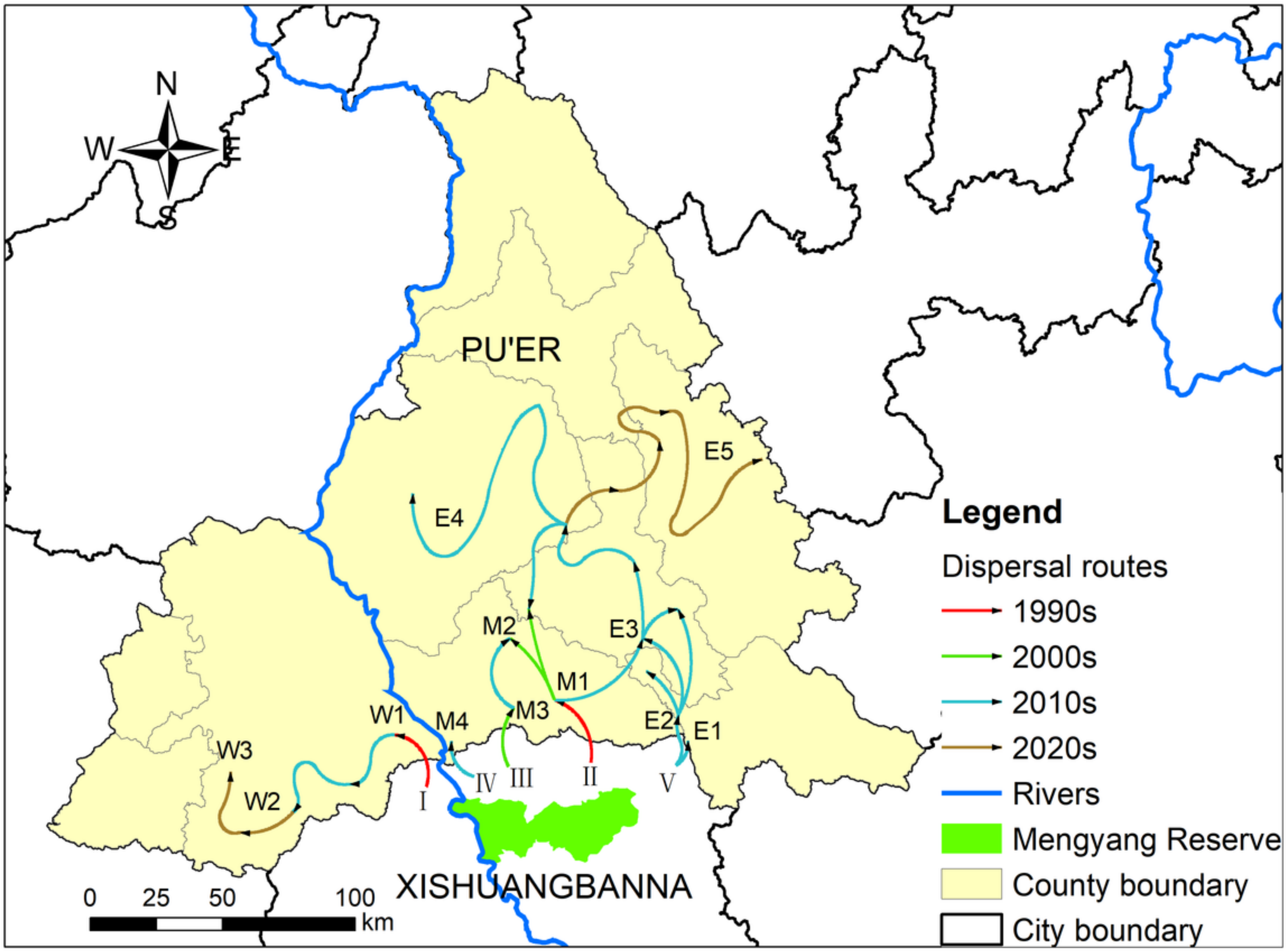

Figure 2

The historical dispersal of the Asian elephants in Pu’er (**W**: the western part of Pu’er, **M**: the central part of Pu’er, **E**: the eastern part of Pu’er; ▯-▯: the herds of the Pu'er-Banna Asian elephant population)

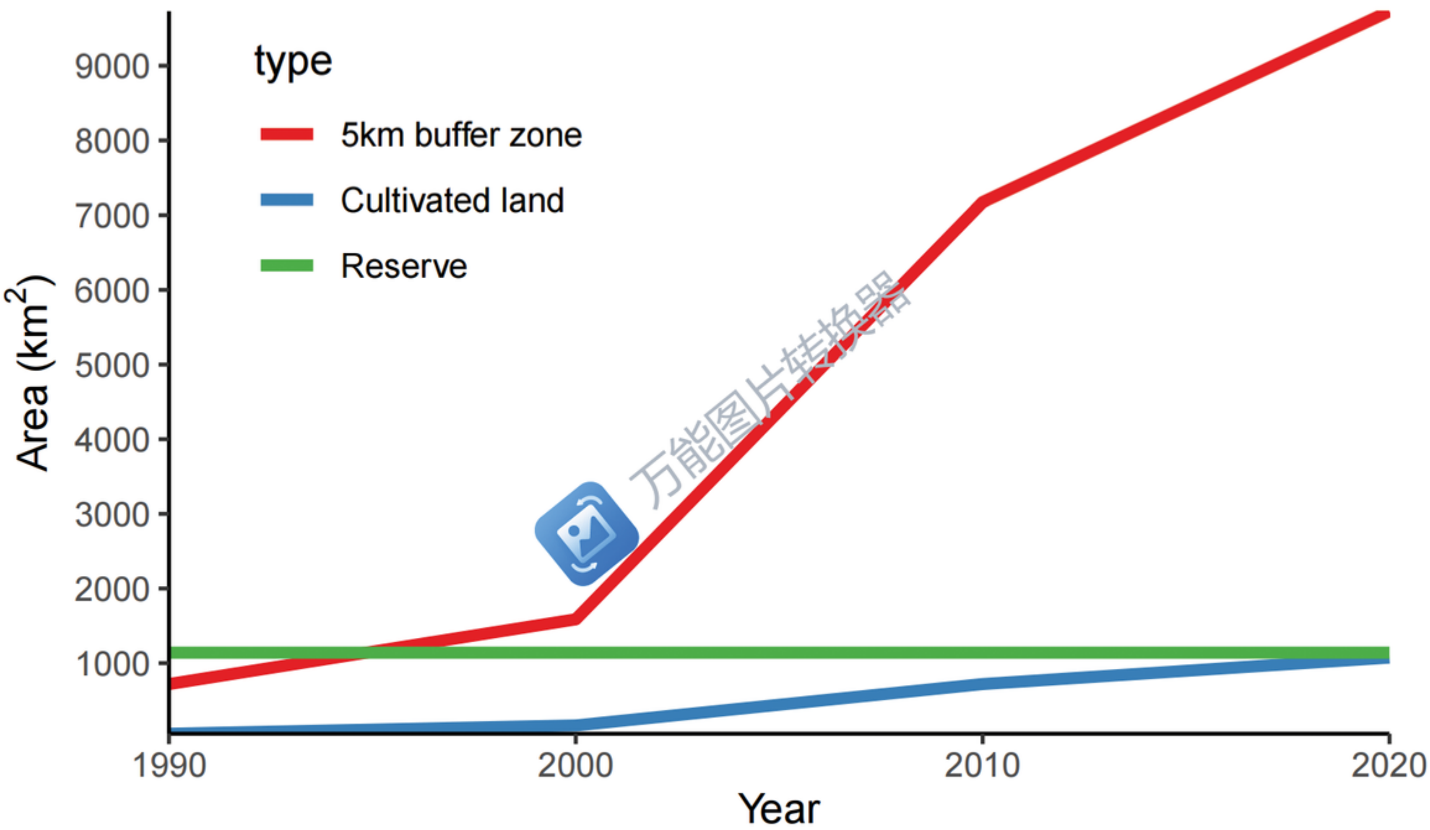

Figure 3

The changes in the area of cultivated land in the 5 km buffer zone along the Asian elephant dispersal route in four periods

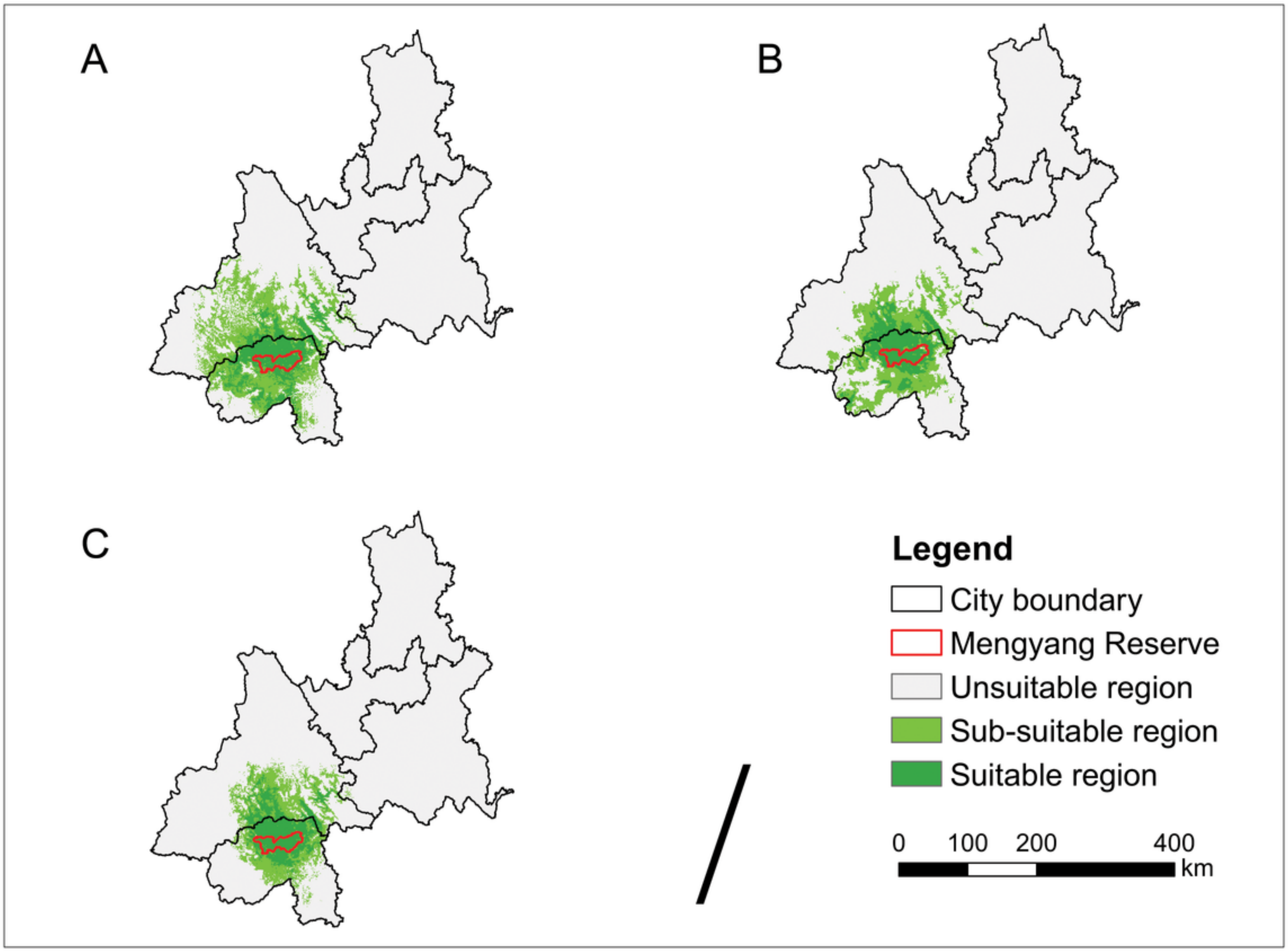

Figure 4

Distribution of Asian elephant distribution range divided by climate in the past (**A**), present (**B**), and future (**C**)

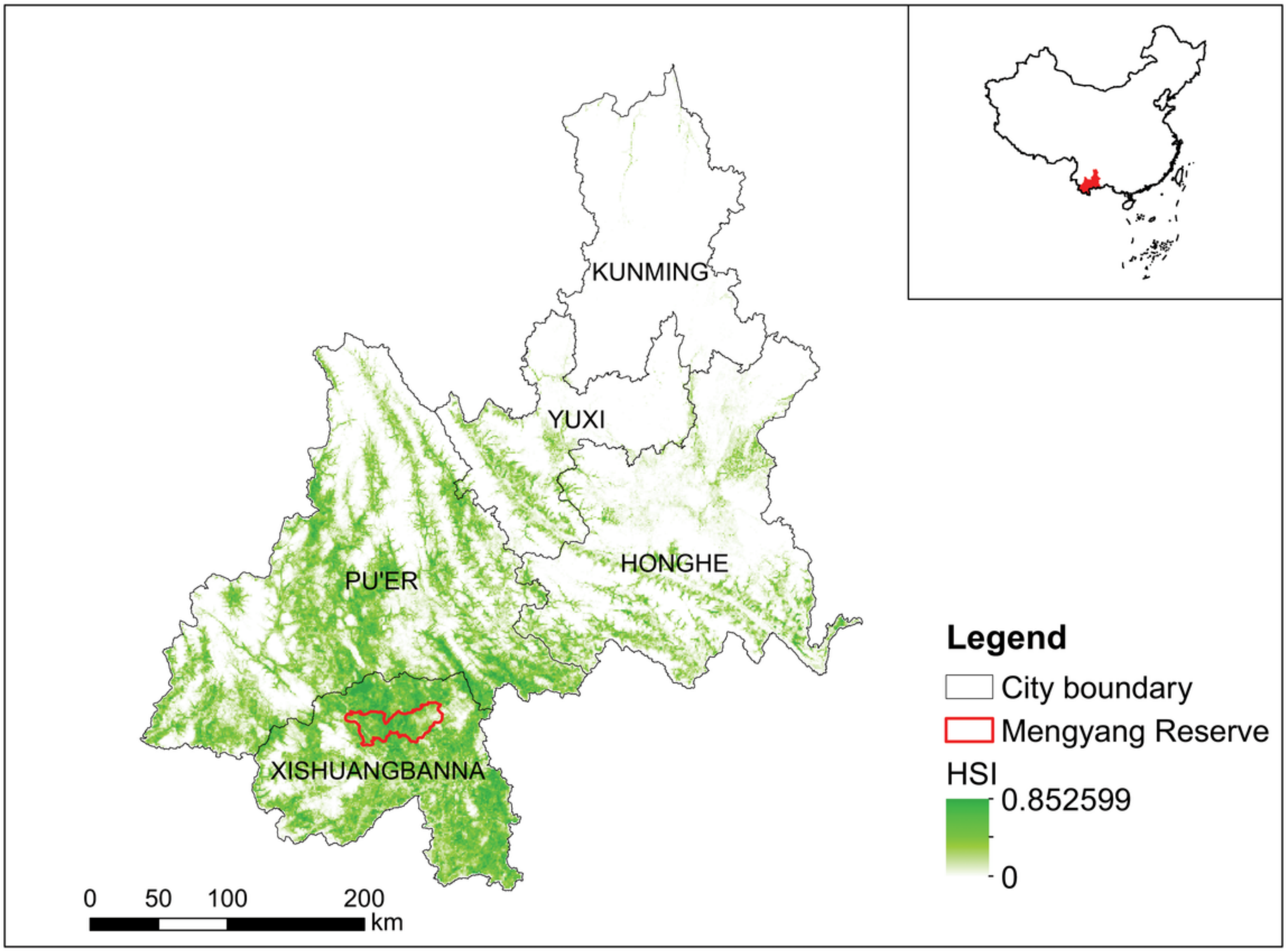

Figure 5

Current habitat suitability of Asian elephants in the study area

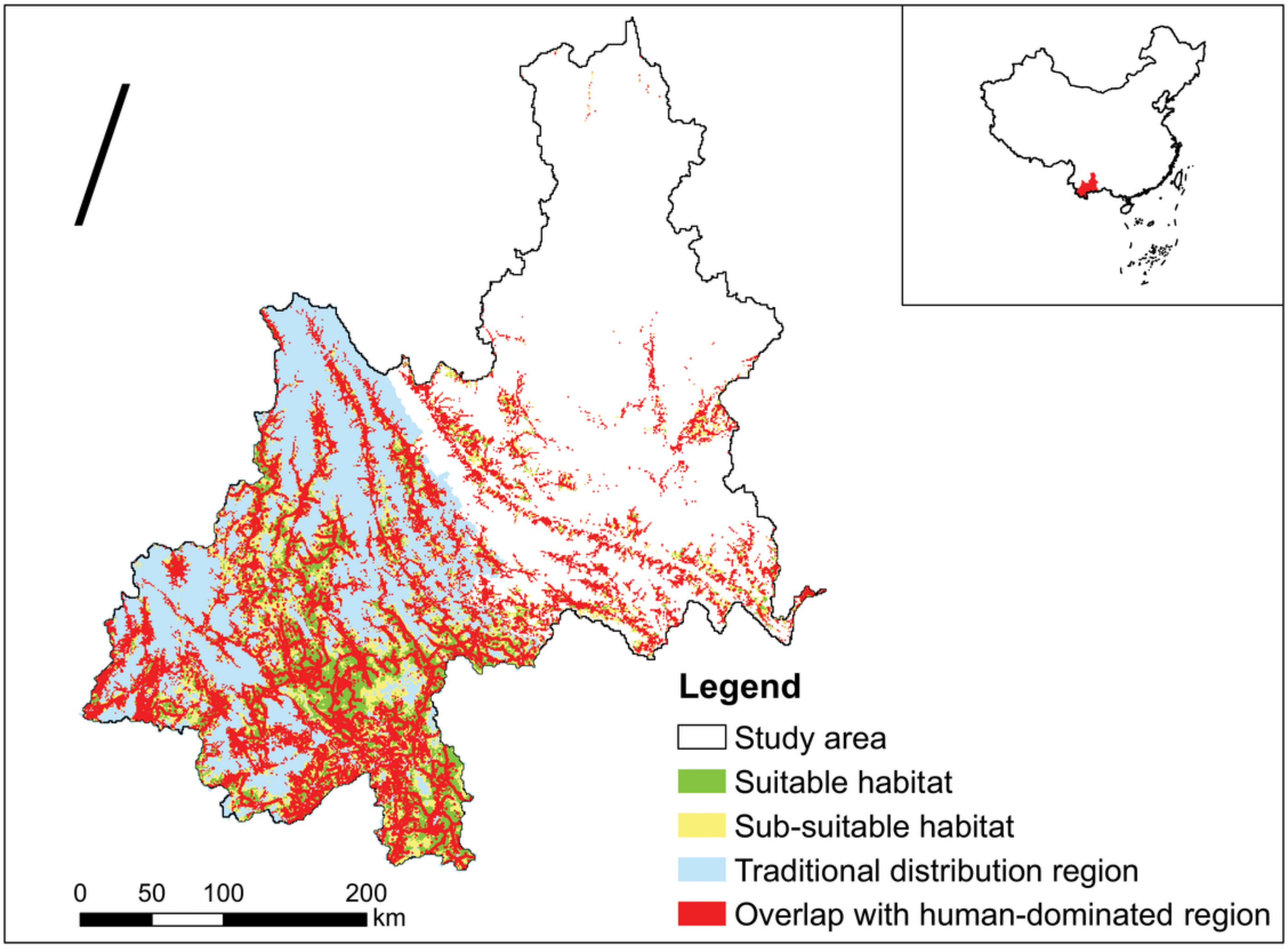

Figure 6

The overlap between the habitat of Asian elephants and the human-dominated region in the study area

Supplementary Files

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